TAO PeopleSemSegNet person semantic segmentation

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> Applicable environment: Jetson Orin NX, JetPack 7.2.1, DeepStream 9.1.

1. Case goal and processing flow

The semantic segmentation predicts a category for every pixel. This model divides the pixels into person/background; it does not distinguish the instances of adjacent people and it does not output an identity ID.

```
Video/USB/CSI → the PeopleSemSegNet mask → person/background
     → the semi-transparent overlay of the original image → the desktop/VNC preview
```

2. The checks before running it

```
cd ~/deepstream_tao_apps/deepstream_app_tao_configs

# Confirm the configuration and the input source
grep -E "uri=|config-file=" run_peoplesem.txt

# Confirm the model and the parser
ls -la /opt/nvidia/deepstream/deepstream-
9.1/samples/models/tao_pretrained/peoplesemsegnet_vdeployable_shuffleseg_unet_onnx_v1.0.1/
ls -la ~/deepstream_tao_apps/post_processor/libnvds_infercustomparser_tao.so
```

Operate in the graphical session of the Jetson desktop or of the VNC. The video mode checks that the material exists; the USB/CSI mode uses the real-time camera entry that is already connected.

3. Start the detection

The video mode:

```
export DISPLAY=:0 && deepstream-app -c
~/deepstream_tao_apps/deepstream_app_tao_configs/run_peoplesem.txt
```



The application loads PeopleSemSegNet and the self-written `NvDsInferParseCustomPeopleSemSegNetTAO` parser (the parser is compiled in `~/deepstream_tao_apps/post_processor/`), the `[segvisual]` section is enabled in the configuration, and the mask is overlaid in the preview:

The current display mode of `[sink0]` only previews. When the natural playback ends, wait for the process to exit; to stop early, press `Ctrl+C` once and wait for the pipeline to wind up.

To replace the test video: modify the `uri=file://` of the `[source0]` section in `run_peoplesem.txt` to the absolute path of an MP4/H.264 file on the Jetson itself; confirm that the Jetson can decode it before you run it. Keep the original material and avoid overwriting something that cannot be restored directly.

4. The real-time detection with a USB camera

The USB mode supports the real-time preview. After confirming that the USB camera device node exists, run:

```
ls -l /dev/video*
v4l2-ctl -d /dev/video0 --list-formats-ext
export DISPLAY=:0 && ~/deepstream_tao_demo/run_case.sh peoplesem usb
```



The script detects the USB camera node automatically through `~/deepstream_tao_demo/find_usb_cam.sh`.

The USB mode is a real-time preview and it does not produce an offline MP4. The input resolution is written dynamically by the script according to the capability of the camera,
and the display is unified with the video mode as the semantic segmentation color effect of `960×544`; it does not show the detection boxes or the text.

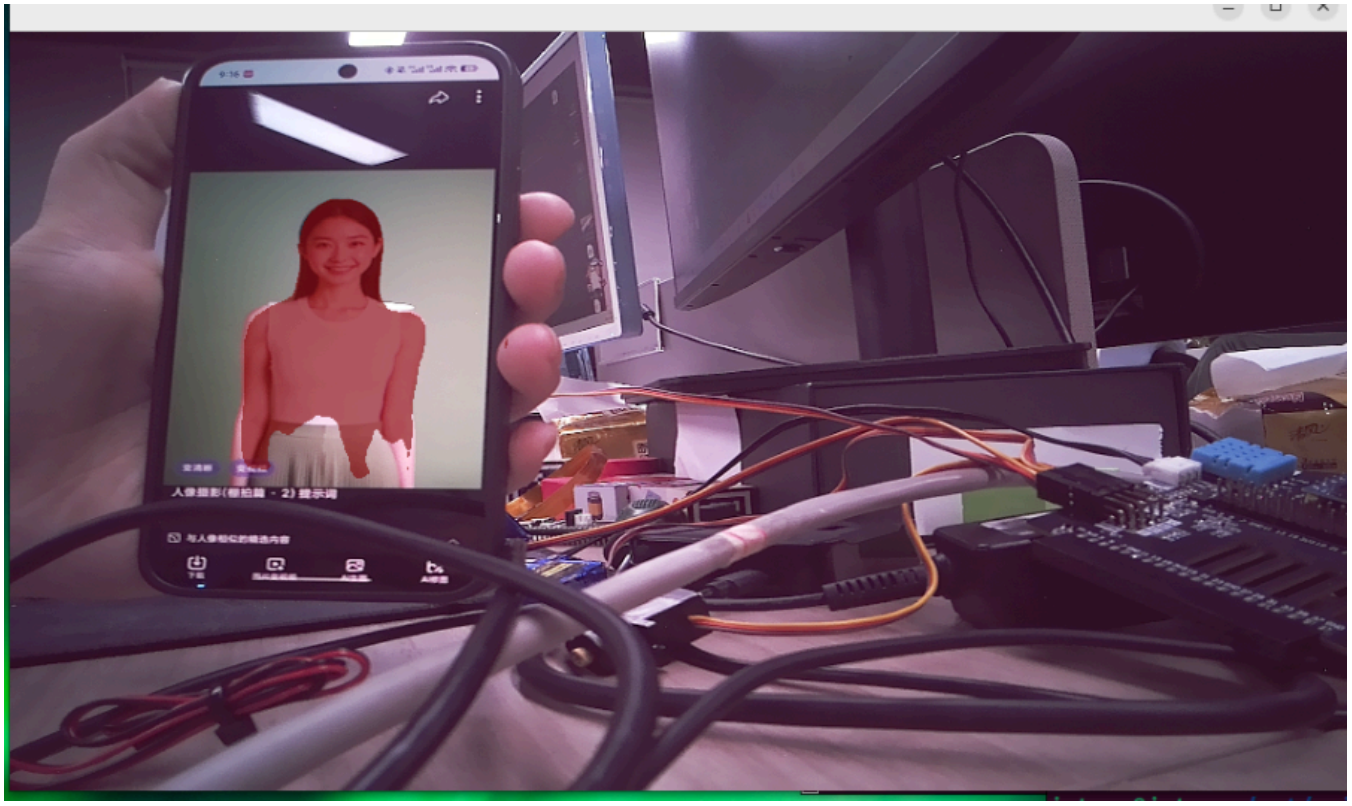
Press `Ctrl+C` once and wait for the application to exit and for the camera to be released.

5. The real-time detection with a CSI camera

The CSI mode supports the real-time preview, and the entry captures through the Argus.

Start:

```
export DISPLAY=:0 && ~/deepstream_tao_demo/run_case.sh peoplesem csi
```



The USB/CSI is a continuous stream and it does not naturally play to the end. To stop it, press `Ctrl+C` once in the terminal that started it and wait for the application and the camera relay to exit before you start another camera task.

6. Frequently asked questions

Symptom	The check and the handling
There is no preview	Check the graphical session, the input path, the model configuration and the compiled artifact of the parser.
The segmented region is missed or it flickers	It is affected by the viewing angle, the occlusion, the size of the person and the domain of the training data; the opacity only changes the display effect.
`Failed to load config file`	Check the `config-file=` path in `run_peoplesem.txt` (the trap of the path missing the `configs/` prefix was hit before); confirm that it points to `~/deepstream_tao_apps/configs/nvinfer/peoplesemsegnet_tao/pgie_peoplesemsegnet_tao_config.txt`.
It reports that `NvDsInferParseCustomPeopleSemSegNetTAO` cannot be found	Confirm that the function is in `~/deepstream_tao_apps/post_processor/libnvds_infercustomparser_tao.so`: `nm -D ... | grep PeopleSemSegNetTAO`.
The `[segvisual]` section does not take effect	That section only supports 7 keys (`enable`, `batch-size`, `gpu-id`, `width`, `height`, `nvbuf-memory-type`, `unique-id`); the extra keys cause the parsing to fail.
The USB/CSI differs from the result of the tutorial	The three configurations uniformly use `960×544`, `segvisual`, `display-text=0` and `display-bbox=0`, and they do not use the Tracker.
There is no image with USB/CSI	Check the device node, the format, the permissions and the occupancy; the CSI entry directly uses the Argus `type=5`.
`nvv4l2h264enc has no property named 'gpu-id'`	It is a known non-blocking warning of DS 9.1 and it does not affect the inference or the display.

7. References

- The NVIDIA DeepStream TAO application examples: https://github.com/NVIDIA/DeepStream/tree/main/src/apps/tao_apps
- The DeepStream TAO integration documentation: https://docs.nvidia.com/metropolis/deepstream/dev-guide/text/DS_TAO_integration.html